6.3 Pascal's Triangle – Worksheet #2

MCR3U

1) Expand each expression using Pascal's triangle

a) $(x + 4)^3$

b) $(1 - 2x)^4$

c) $(3x + y)^2$

d) $(x - 5)^5$

e) $(3 + 2n)^6$

f) $(x - 7)^{11}$

g) $(2x + 5)^7$

Answers

1. a) $x^3 + 12x^2 + 48x + 64$
b) $1 - 8x + 24x^2 - 32x^3 + 16x^4$
c) $9x^2 + 6xy + y^2$
d) $x^5 - 25x^4 + 250x^3 - 1250x^2 + 3125x - 3125$

e) $729 + 2916n + 4860n^2 + 4320n^3 + 2160n^4 + 576n^5 + 64n^6$

f) $x^{11} - 77x^{10} + 2695\,x^9 - 56\,595x^8 + 792\,330x^7 - 7\,764\,834x^6 + 54\,353\,838x^5 - 271\,769\,190\,x^4 + 951\,192\,165\,x^3 - 2\,219\,448\,285\,x^2 + 3\,107\,227\,739x - 1\,977\,326\,743$

g) $128\,x^7 + 2240\,x^6 + 16\,800\,x^5 + 70\,000\,x^4 + 175\,000\,x^3 + 262\,500\,x^2 + 218\,750\,x + 78\,125$